

Understanding the Contributions of the Milky Way and M31 to the Merger Remnant Shape (Research Assignment 2)

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1. INTRODUCTION

The area of galaxy evolution this research will investigate is the impact galaxy mergers have on dark matter halos. Dark matter is believed to be dispersed throughout the universe in an almost web-like structure with a vast system of filaments. Dark matter halos are the areas in the galaxy where there is a high density of dark matter. These halos are theorized to come about when multiple filaments intersect with one another. These dark matter halos are significant because they are where galaxies are able to form [C. Frenk & S. White \(2012\)](#).

Because these dark matter halos are the basic unit of cosmological structure, understanding dark matter halos is intrinsic to understanding cosmology. Researching the structure of dark matter halos, as well as halo formation and halo growth can give key insights into galaxy evolution and formation. Understanding halo growth requires understanding how these structures are impacted when galaxies merge.

The main understanding of dark matter halo mergers comes from cosmological simulations of mergers set in near perfect settings. So far the general understanding finds that the size and spin of the halo remnant are dependent on the energy of the orbit and the angular momentum of the orbit respectively. The shape of the remnants is primarily dependent on the initial separation between the two halos as well as the spin of the mergers [N. E. Drakos et al. \(2019a\)](#), as seen in Figure 1. Although the relationship between energy of a system and the spread (or how compacted/extended) of a remnant appears to be direct, the correlation between the energy of a system and the radius of the scaled density profile of the remnant appears to be an inverse relationship [N. E. Drakos et al. \(2019b\)](#).

A lot of the understanding of dark matter halo mergers comes from simulations done with equal mass binary systems, which is a simplified setting for mergers, as it does not take into account mergers between different sized halos or mergers between more than two halos. These simulations do not take into account halo substructures [N. E. Drakos et al. \(2019a\)](#). One of the largest issues still being examined from these simulations is what causes the central density of merger remnants to decrease, because the current simulations see an increase in central density that does not match the drop seen in central density of halos as they grow [N. E. Drakos et al. \(2019b\)](#). Other questions left open include how the relationships and general findings of these merger simulations might change when including more complicated and more common types of mergers, such as mergers between more than two halos.

2. PROPOSAL

2.1. 1. *This Proposal*

The specific question that will be addressed in this research assignment is what the particles from the Milky Way halo and the M31 halo contribute to the final shape of the remnant. This research will look at where the specific halo particles from each galaxy end up coalescing in the final remnant. Furthermore, this work will also investigate the ratio of intermixing occurs between particles from each galaxy, as well as if different regions of the final remnant are dominated by the particles of a specific original galaxy.

2.2. 2. *Methods*

The particle types this research will be using for analysis are the Type 1 dark matter or halo particles. This research will be looking at snapshots after the two galaxies merge, so in roughly 6.5 Gyr, which correlates to snapshots 435 and beyond. This research will compare multiple snapshots after merger, including snapshot 435, 535,

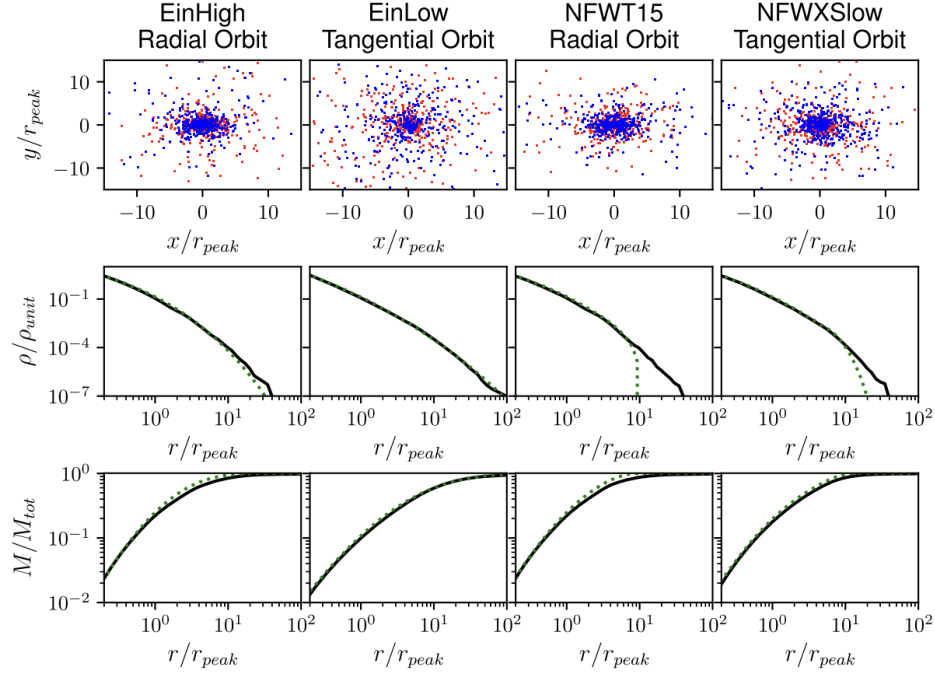
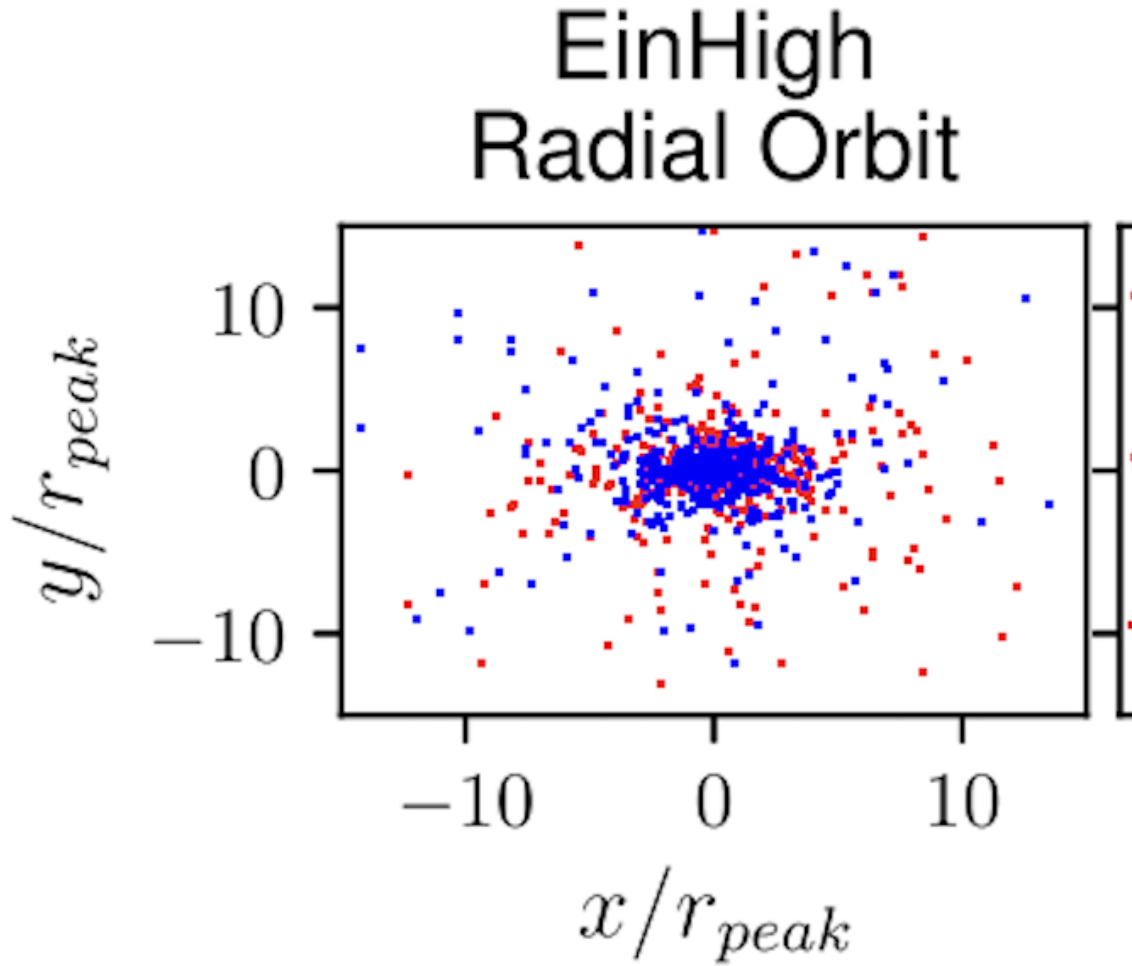


Figure 10. Sample results from four merger simulations. The top row of panels shows the state of the remnant at $t = 300 t_{\text{unit}}$ (in this plot we show only 10^3 randomly selected particles). The particles are coloured either red or blue, depending on their initial halo. The middle row of panels shows the density profiles, and the bottom row shows the mass profiles. The scaled ICs and remnants are shown with dotted green lines and solid black lines, respectively. All haloes were initially separated by $r_{\text{sep}} = 10$, and given either a radial or tangential velocity of $v_0 = 0.8 v_{\text{esc}}$, where v_{esc} is the escape speed of a point mass in the potential of the initial halo.

Figure 1. Figure 10 from Drakos + MNRAS, 2019a, 487, 993.

635, and 735 to compare how the shape of the remnant as well as the dispersion of the particles from the two original galaxies, changes over time. The way this research will be addressed is by creating a 3D model of the remnant and color coding the particles that make up the remnant according to where they originally came from. This will be done by plotting the particles (or a random sample of particles) in their x, y, and z coordinates. This model will give a visualization of the particle makeup of the remnant, but for further analysis a few different follow up treatments may be performed depending on the outcome of the 3-D structure. One analysis may be a comparison of the ratio of Milky Way particles to M31 particles in shells of varying distances from the center of the remnant. This would help see what the intermixing of particles looks like at different distances from the origin. If the 3-D model is not uniform, or if there are visually distinct clusters of particles from particular galaxy origins, another form of analysis may involve dividing the remnant into 8 distinct sectors, and performing mass analysis by looking at the ratio of particles in the portion of the shells in that sector. The extent of the analysis performed will be determined in full after the initial 3-D structure is created and analyzed. This 3-D model as well as analysis of the different particle origins will be made for each snapshot to see if there are major changes in the percentages over time. A model similar to this one but in three dimensions will be expected for the Research:



N. E. Drakos

et al. (2019a)

2.3. 3. Hypothesis

Due to the path of collision the Milky Way and M31 are on, the expected outcome of this research is to find a 3D simulation with the qualities similar to the radial orbit remnants shown in Figure 10 from Drakos et al. The anticipated end result will have particles from each of the galaxies intermixing, but with one galaxy's particle type dominating the center of the remnant while the other galaxy's particle types become more prominent the further you get from the center of the remnant.

REFERENCES

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